

# Karen Shih

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## EDUCATION

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### Stanford University — PhD Candidate, Biology

2020 – Dec 2026 (expected)

**Research advisors:** Professors Dmitri Petrov & Monte Winslow

**Thesis committee:** Professors Dmitri Petrov, Monte Winslow, Onn Brandman, Hunter Fraser

**Dissertation:** Pan-cancer transcriptomics and *in vivo* functional screens to assess tumor suppression in high tumor mutational burden cancers

### University of California, Berkeley — BA, Molecular & Cell Biology

2014 – 2018

**Undergraduate thesis advisor:** Professor Michael Rape

**Undergraduate honors thesis:** Uncovering the redox control of a ubiquitin ligase

## RESEARCH EXPERIENCES

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- **Pan-cancer TMB transcriptional response** — sole first author, computational
  - Ran generalized linear mixed effect models (GLMM) on pan-cancer TCGA data (~10,000 tumors) to characterize the per-gene association between tumor mutational burden (TMB) and gene expression across 32 cancer types (~16,000 genes analyzed)
  - Constructed a TMB axis score as a weighted sum of gene expression values using gene-wise GLMM  $\beta$  coefficients to summarize the sample-level response to increasing TMB
  - Implemented bootstrapping, leave-one-out cross-validation by cancer type, and 70/30 training/validation split to assess robustness of pan-cancer TMB-associated transcriptional response
- **High-TMB cancer models & *in vivo* CRISPR screens** — sole first author, computational + experimental
  - Developed two novel within-mouse-control systems for high-TMB cancers: hygromycin-induced translational infidelity and aggregation-prone protein overexpression
  - Conducted a power analysis with simulations of high TMB effects and gene knockout effects to optimize CRISPR screen design, determining the dual guide screen scale for the translational infidelity mouse model: 121 genes (60 tumor suppressors, 61 proteostasis genes) across 50 mice at 80% power
  - Designed and executed *in vivo* CRISPR knockout screens across ~120 mice (30,000-50,000 tumors/mouse) for the two models to identify tumor suppressor genes and context-dependent essential genes in high-TMB tumors
  - Executed *in vivo* experiments: viral vector production, drug treatment, tissue collection and processing
  - Validated preliminarily the translational infidelity mouse model through RNA-seq analysis, assessing parallels of misfolded protein stress in human high-TMB tumors and the high-TMB mouse model
- **Fusion oncogene & EML4-ALK collaborations** — co-first author, computational
  - Developed quality control frameworks to detect and filter multiple infections, sample contamination, and non-dividing inert controls to ensure per-gene effect estimates reflect genuine tumors rather than artifacts
  - Quantified seven EML4-ALK variants' oncogenic and metastatic potential from the *in vivo* screen data
  - Advised the screen design by recommending guides that reliably form tumors and treatment conditions to increase power to detect hits

## PUBLICATIONS IN PREPARATION

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1. Siah VP\*, **Shih KY\***, Xu H, Andrejka L, Hebert JD, Winslow MM. Somatic genome editing initiates diverse, oncogenic fusion-driven lung adenocarcinoma tumors in Cas12a mice. In preparation. \*Equal contribution.
2. **Shih KY**, Andrejka L, Brandman O, Winslow MM, Petrov DA. The fitness landscape of tumor suppression and synthetic lethal genes in a high tumor mutational burden mouse model. In preparation.
3. Najajreh M\*, **Shih KY\***, Lee D, Manzano S, Ramos M, Somogyi K, Poth T, Kinas L, Kronemeijer H, Karandikar S, Razic E, Gorkuslar C, Diaz-Jimenez A, Lu J, Scherer M, Winslow MM, Sotillo R. EML4-ALK variants shape metastatic progression and response to therapy. In preparation. \*Equal contribution.

## PEER REVIEWED PUBLICATIONS

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1. **Shih KY**, Brandman O, Winslow MM, Petrov DA. Transcriptional response to tumor mutational burden is consistent across cancer types. *BioRxiv*. 2026. DOI:10.64898/2026.08.27.744035.
2. Diaz-Jimenez A\*, Shuldiner EG\*, Somogyi K, **Shih KY**, González-Velasco Ó, Najajreh M, Kim S, Akkas F, Murray CW, Andrejka L, Tsai MK, Brors B, Hofmann I, Sivakumar S, Sisoudiya SD, Sokol ES, Cai H, Petrov DA, Winslow MM, Sotillo R. EML4-ALK variant-specific genetic interactions shape lung tumorigenesis. *Cancer Discovery*. 2025. DOI: 10.1158/2159-8290.CD-24-1417. \*Equal contribution.
3. McMinimy R, Manford AG, Gee CL, Chandrasekhar S, Mousa GA, Chuang J, Phu L, **Shih KY**, Rose CM, Kuriyan J, Bingol B, Rapé M. Reactive oxygen species control protein degradation at the mitochondrial import gate. *Molecular Cell*. 2024. DOI: 10.1016/j.molcel.2024.11.004.
4. Manford AG\*, Mena EL\*, **Shih KY**, Gee CL, McMinimy R, Martínez-González B, Sherriff R, Lew B, Zoltek M, Rodríguez-Pérez F, Woldesenbet M, Kuriyan J, Rape M. Structural basis and regulation of the reductive stress response. *Cell*. 2021. DOI: 10.1016/j.cell.2021.09.002. \*Equal contribution.
5. Manford AG, Rodríguez-Pérez F, **Shih KY**, Shi Z, Berdan CA, Choe M, Titov DV, Nomura DK, Rape M. A Cellular Mechanism to Detect and Alleviate Reductive Stress. *Cell*. 2020. DOI: 10.1016/j.cell.2020.08.034.

## POSTER PRESENTATIONS

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1. The Fitness Landscape of Tumor Suppression in High Tumor Mutational Burden Tumors. Mechanisms and Models of Cancer, 2024
2. The Fitness Landscape of Tumor Suppression in High Tumor Mutational Burden Tumors. CMB Program Symposium, Stanford University, 2023
3. The Fitness Landscape of Tumor Suppression in High Tumor Mutational Burden Tumors. Biology Department Retreat, Stanford University, 2023
4. The Fitness Landscape of Tumor Suppression in High Tumor Mutational Burden Tumors. Biology Department Retreat, Stanford University, 2022

## AWARDS & FELLOWSHIPS

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**Stanford Community Impact Award**, Stanford University, 2023

- Recognizes graduate students for outstanding leadership and community building through student organizations or campus initiatives

**Eco-Evo Travel Award**, Stanford University, 2022

- Conference travel funding awarded by the Ecology & Evolution track within the Biology Department

**Cellular and Molecular Biology Program Award**, Stanford University, 2021 – 2023

- Competitive fellowship covering stipend and tuition for two years of doctoral training

## TEACHING EXPERIENCES

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**Teaching Assistant**, Stanford University

Cell Biology (BIO 86) and Topics in Cancer Biology (BIO 124), 2021

- Led discussion sections for undergraduate and graduate students

## REFERENCES

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**Professor Dmitri Petrov** – [dpetrov@stanford.edu](mailto:dpetrov@stanford.edu) – PhD advisor

Michelle and Kevin Douglas Professor of Biology – Stanford University

**Professor Monte Winslow** – [mwinslow@stanford.edu](mailto:mwinslow@stanford.edu) – PhD advisor

Associate Professor of Genetics and Pathology – Stanford University

**Professor Onn Brandman** – [onn@stanford.edu](mailto:onn@stanford.edu) – Collaborator and thesis committee member

Associate Professor of Biochemistry, Chemical and Systems Biology – Stanford University